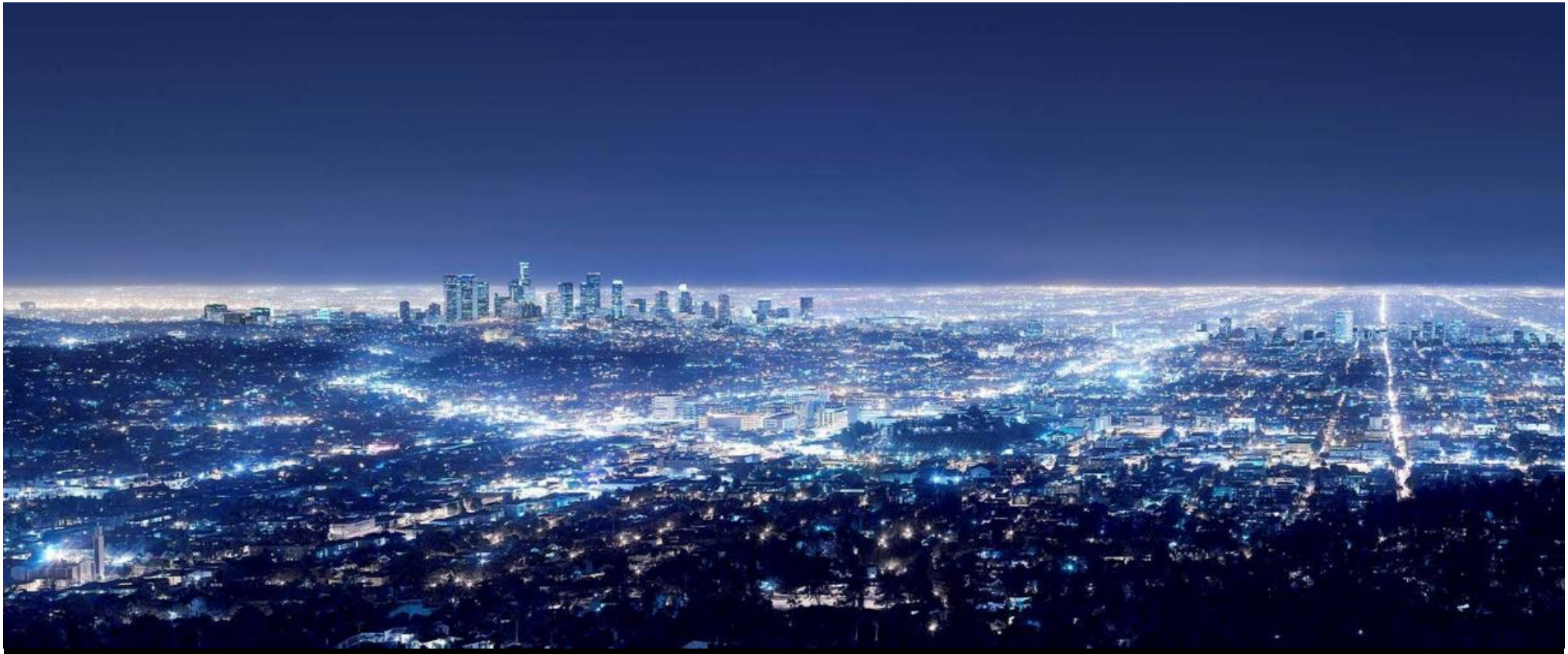


ABB Medium Voltage Products – Sanjay Sharma, 29.11.2016

Substation Communication with IEC 61850 and Application Examples

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Power and productivity for a better world™

Substation communication using IEC 61850
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1. Introduction
2. IEC 61850 Overview, IEC 61850 Edition 2
3. IEC 61850 Horizontal Communication
4. Relion® Performance
5. IEC 61850 in DA products
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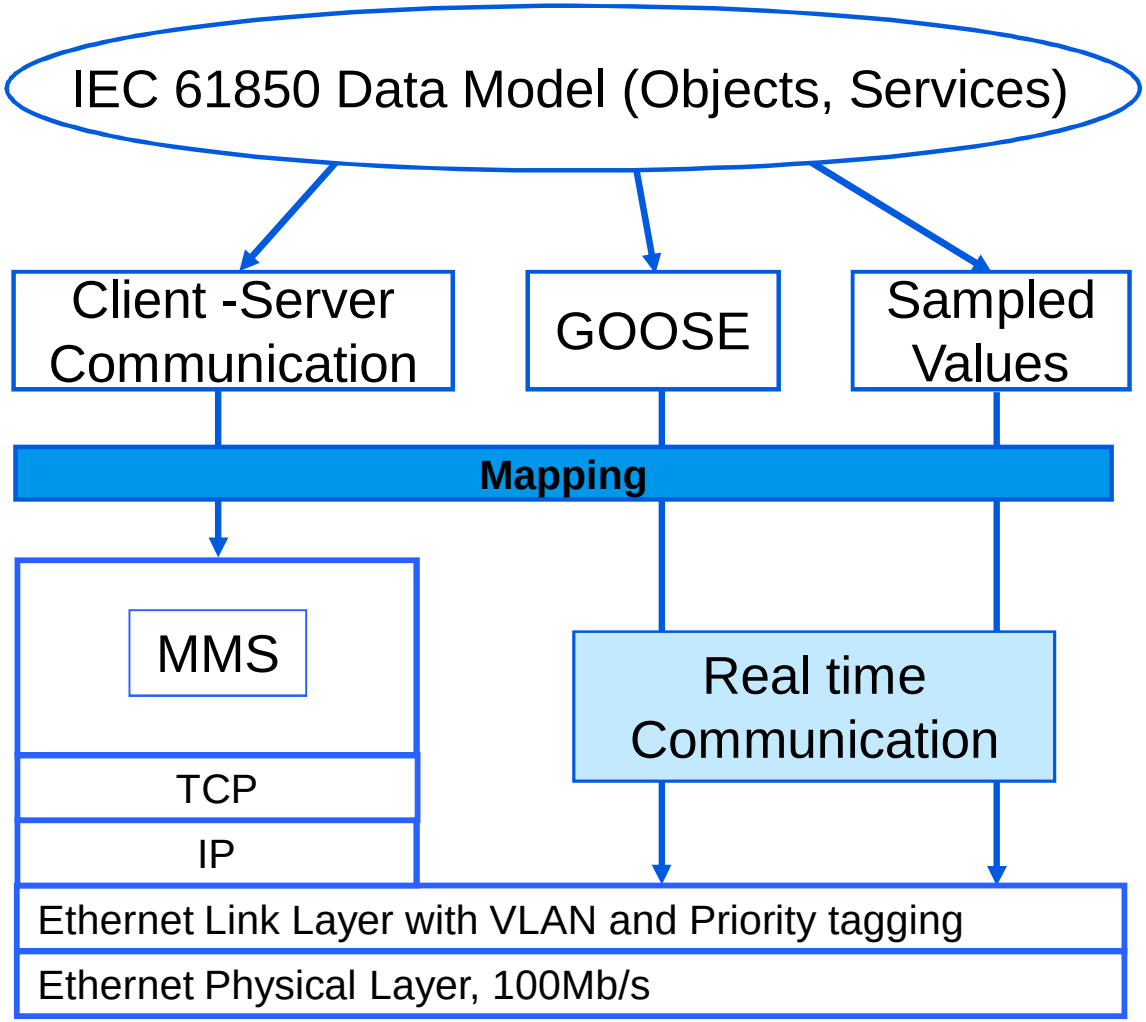
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IEC 61850 Overview
Starting point and set goals



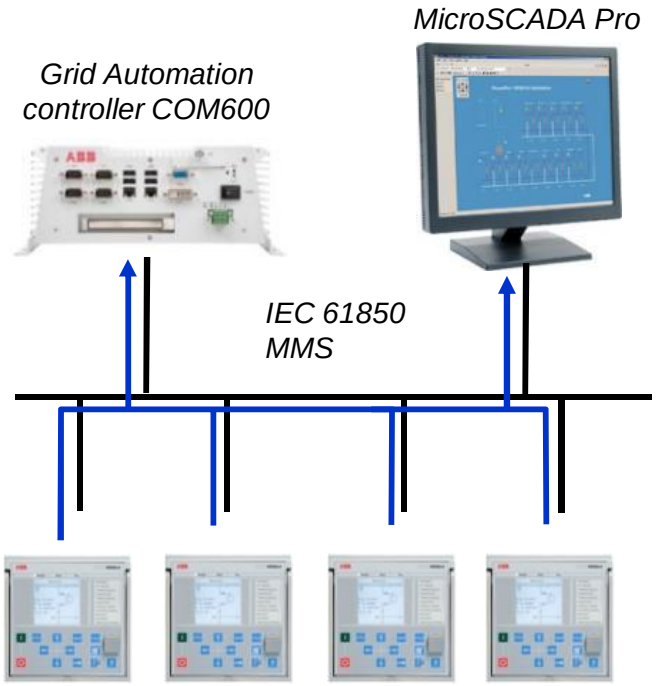
- **Interoperability**
  - The ability of IEDs from one or several manufacturers to **exchange** information and **use** the information for the their own functions**.**
- **Free configuration**
  - The standard shall support different **philosophies** and allow a free allocation of functions e.g. it must work equally well for centralized (RTU like) or decentralized (SCS like) systems
- **Long term stability**
  - The standard shall be **future proof**, i.e. it must be able to follow the progress in **communication technology** and the evolving **system requirements**

IEC 61850 Overview
Communication Profiles



IEC 61850 Overview
Vertical Communication Profile

Introduction
Mechanical design
Module functionality
Communication
Tools
Application examples
Conclusions



- Vertical communication using IEC 61850 MMS profile to COM600, MicroSCADA, RTU560 or 3rd party masters
- Based on normal TCP/IP in Ethernet
- DA relays accepts five parallel IEC 61850 clients
- At the same time also IEC 61850 horizontal communication and other protocols can be used as well

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IEC 61850 Edition 2
Standard continues…

IEC 61850 Edition 1
From proprietary protocols towards a global communication standard **for substation automation**

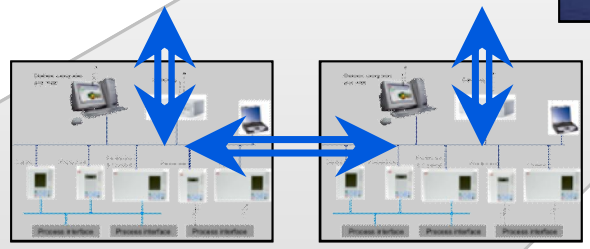
Data model for water and wind application



IEC 61850 Edition 2
A broad, encompassing standard for Communication Networks and Systems **for power utility automation**



TODAY



IEC 61850 Edition 2.1 / 3
New extensions in work for smart grid



IEC 61850
Edition 2

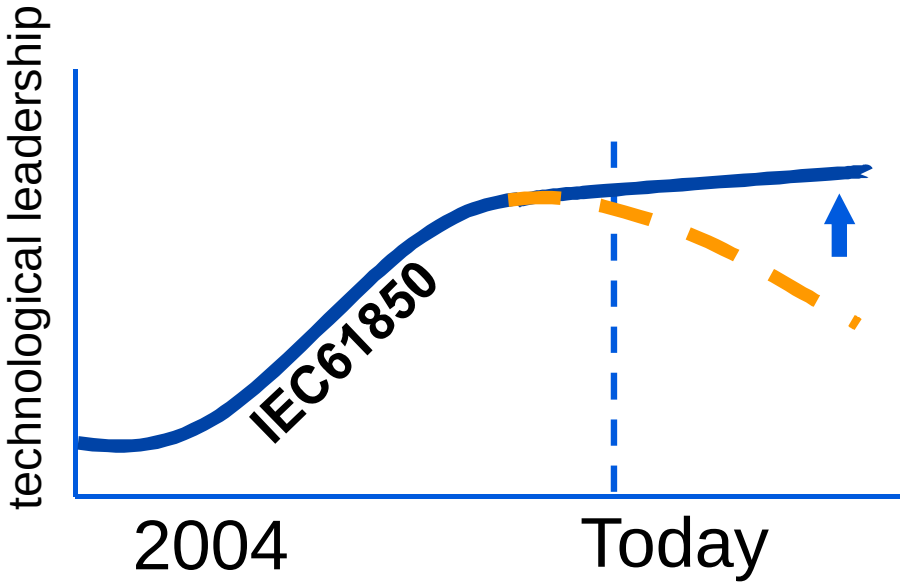


	ABB
Edition 1 certificates	ì
PRP	ì
HSR	ì
Edition 2 certificates	ì

- Technological leadership, Standard is ready
- Market needs and expectations
- Support for both Edition 1 and Edition 2

Edition 2

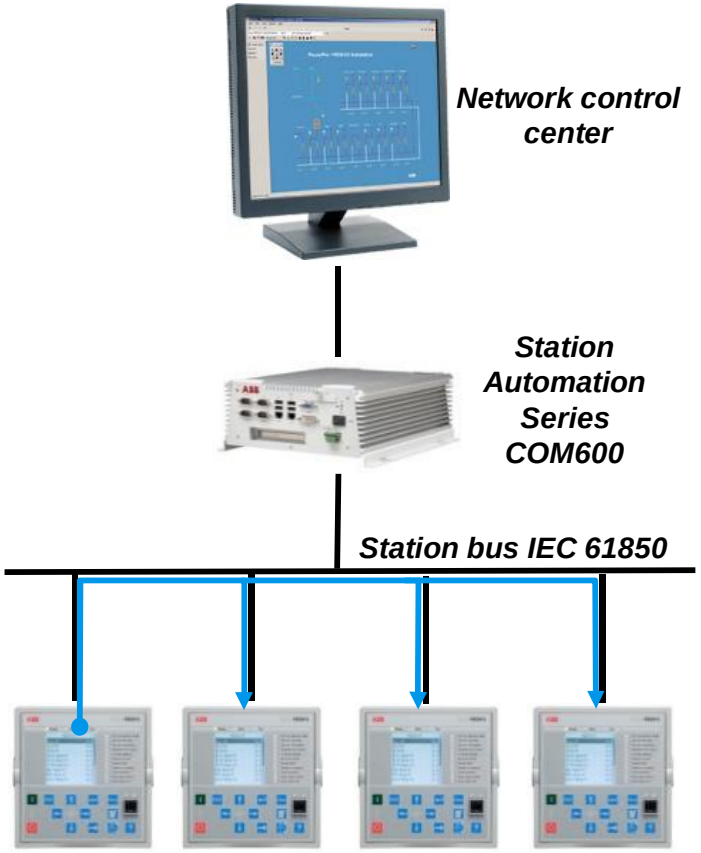
- Mature technology
- More features to support user needs
- Latest version (important life cycle perspective)

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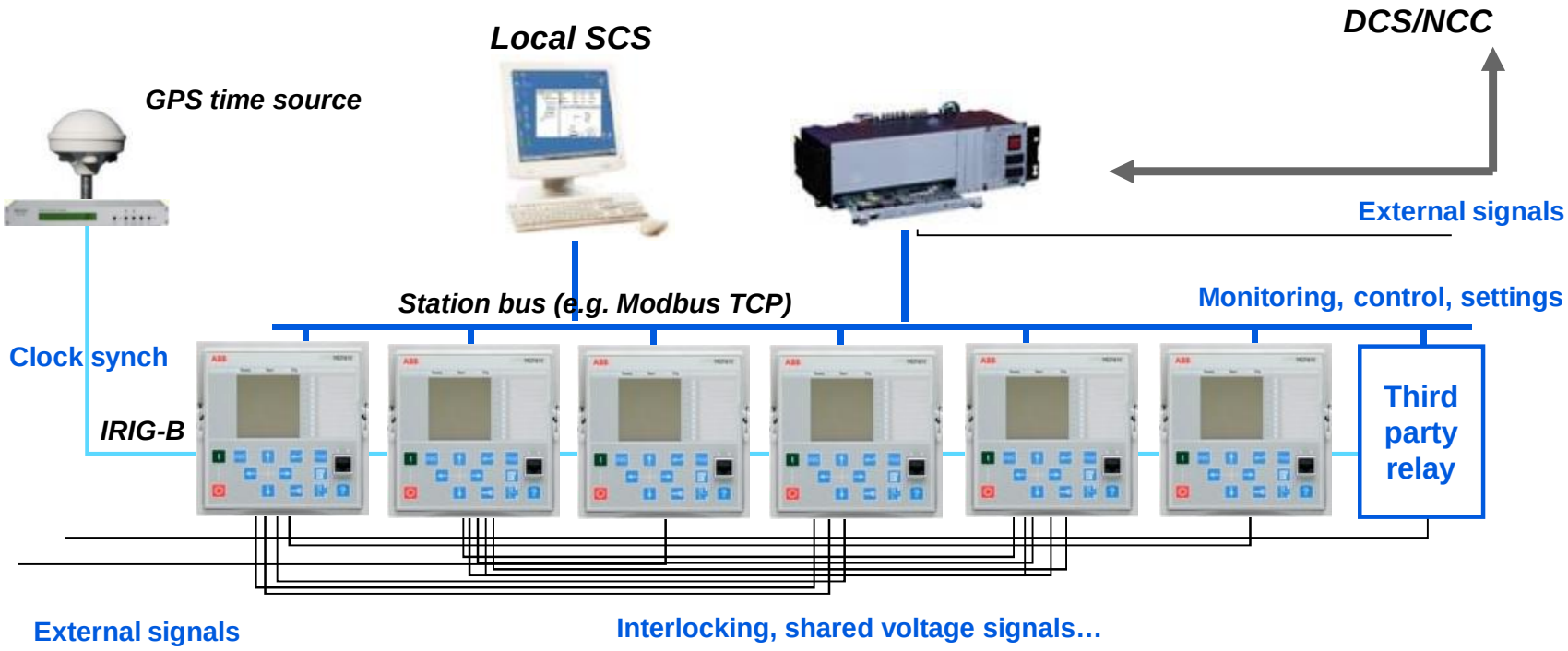
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IEC 61850 Horizontal Communication
GOOSE Communication, General



- Standardized horizontal communication in station bus
  - Ethernet replaces hard-wiring between IEDs
  - GOOSE = Generic Object Oriented Substation Event
- GOOSE is used to broadcast events to peer IEDs in a substation
- Ethernet Multicast messages
- Can in principle transmit any kind of process data between the IEDs
- Ethernet technology offers a fast and reliable station bus for transfer of data
- Supervised communication

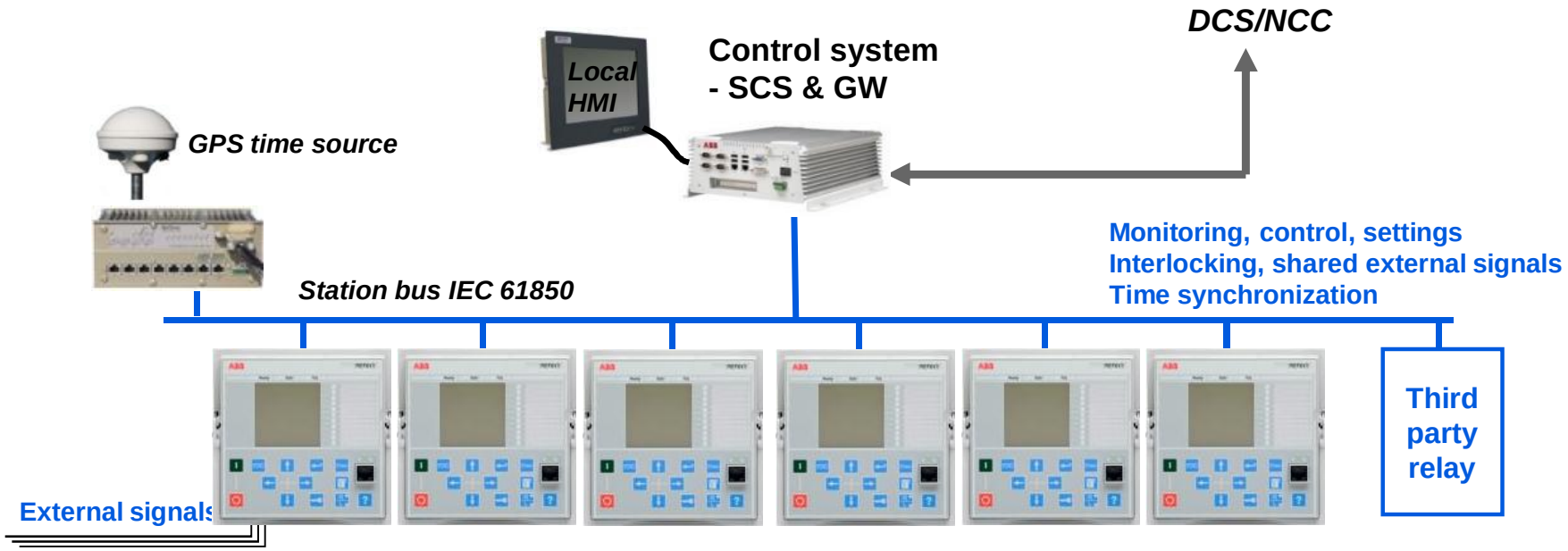
IEC 61850 Horizontal Communication
Traditional Approach



- Conventional copper-wired inter-bay signals like:
  - Time synchronization
  - Protection blocking
  - Control interlocking

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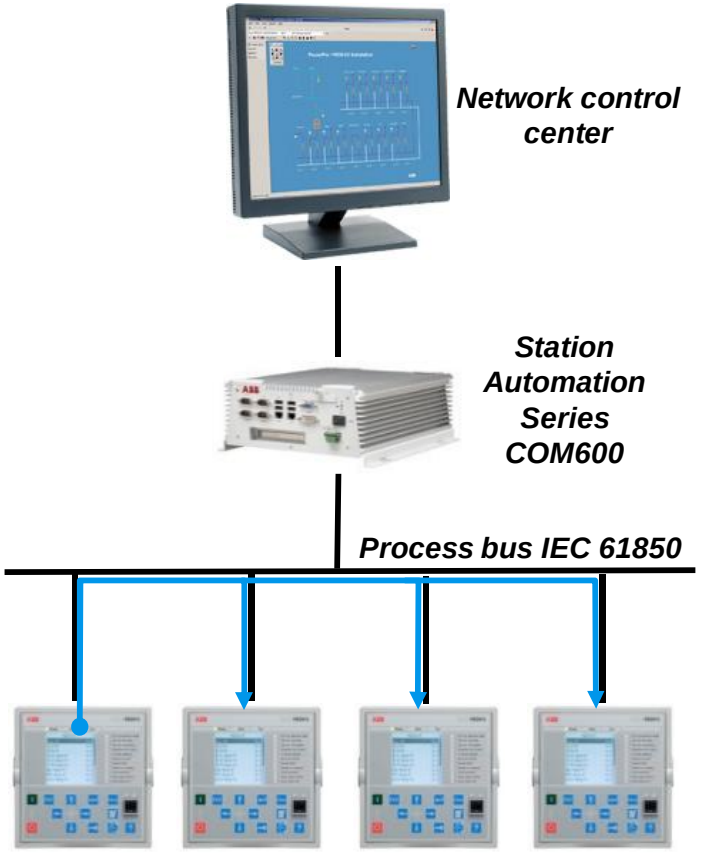
IEC 61850 Horizontal Communication
Station Bus with GOOSE



- Inter-bay signals using IEC 61850 and GOOSE
- SNTP based time synchronization utilising station communication and enabling +/-1 ms event time stamping accuracy

© ABB Group

IEC 61850
SMV Horizontal Communication



- Standardized Process Bus communication in Ethernet
  - Ethernet replaces hard-wiring between IEDs
  - SMV = Sampled Measured Values
- SMV is used to broadcast stream of raw measurement samples to peer IEDs in a substation
- Ethernet Multicast messages with sample rate of 4kHz
- Ethernet technology offers a fast and reliable station bus for transfer of data
- Supervised communication

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IEC 61850 Communication
IEEE 1588 Time Synchronization



- IEEE 1588 (Precision Time Protocol – PTP) version 2 with Power Profile time synchronization method enabling 1 µs high accuracy time synchronization
- Required especially in process bus applications with sampled values using protocol IEC 61850-9-2LE
  - IED has a time-stamp resolution of < 4 µs
- Grand Master Clock sends once per second synchronization message to network as multicast message
- Accurate usage of IEEE 1588 requires hardware level support from all devices in network

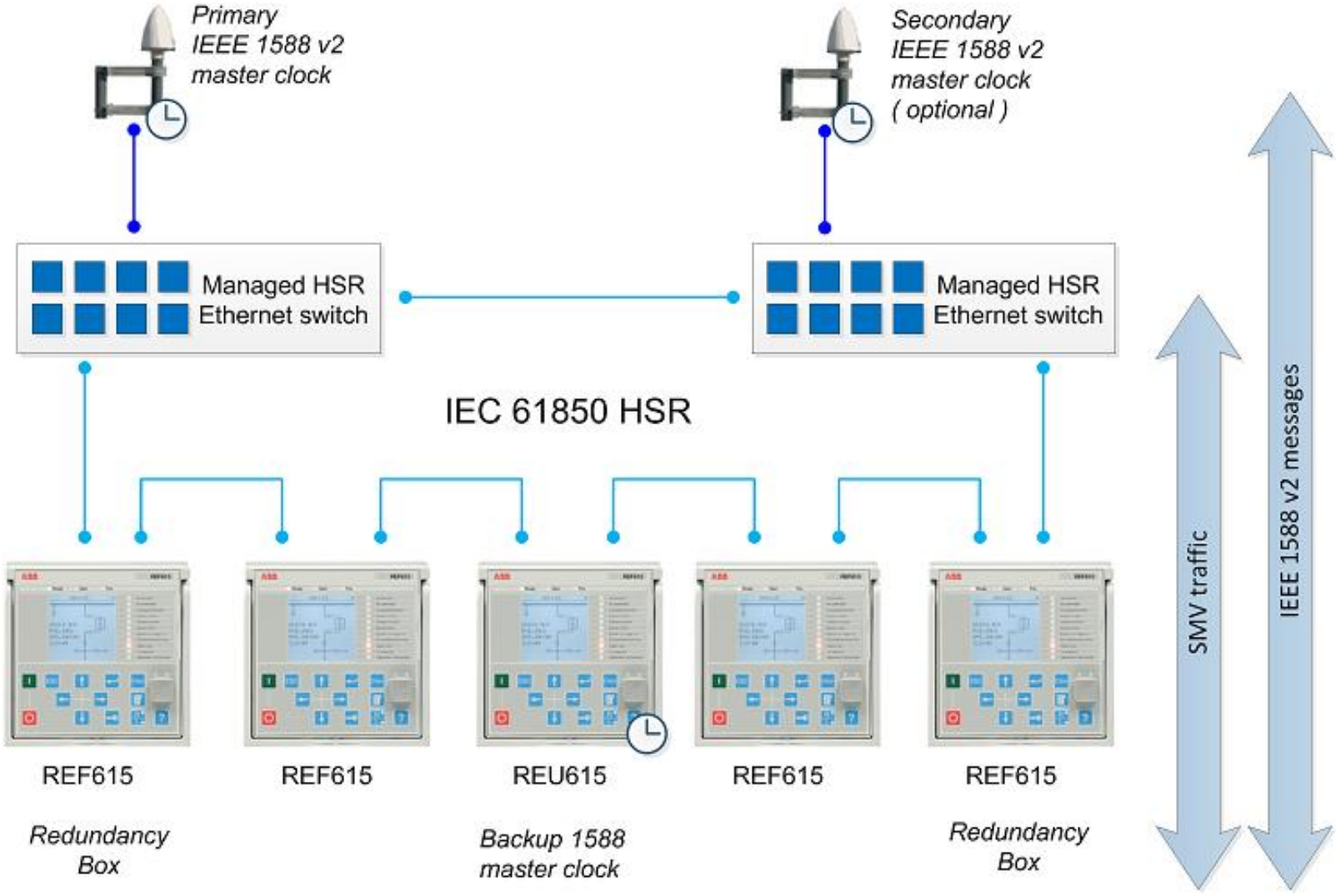
IEC 61850 Communication
IEEE 1588 time synchronization



- Benefits
  - High accuracy (1 µs)
  - Reliable
  - Ethernet and HSR/PRP compliant
- Reliability
  - Redundant clock masters
  - Backup clock masters (IED level)
  - Supervision
- Ready for The most demanding applications
  - 1 µs = 1:1000000s
  - 1 µs < 0.05º

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IEC 61850 Communication
IEEE 1588 Example



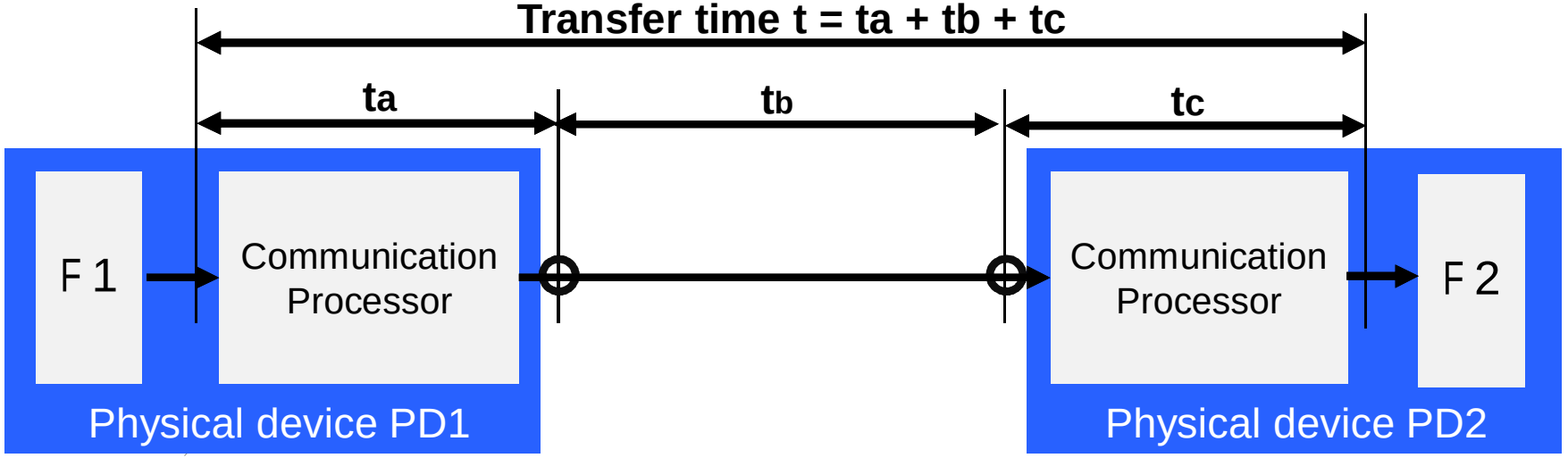
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Relion® Performance
GOOSE Performance Requirements

- GOOSE response time requirements from application to application according the standard (IEC 61850-5) are:
  - Type 1A (tripping)
    - Class P2/3: <3ms
    - **Class P1: <10ms (P1 = distribution S/S)**
  - Type 1B (others)
    - Class P2/3: <20ms
    - Class P1: <100ms



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Relion® Performance
615 and 630 performance testing with KEMA





- Not only the relays are tested for IEC 61850 conformance, but also ABB switchgear solution was tested in Brno switchgear factory high-voltage lab
  - **IEC 62271-3, High-voltage switchgear and control gear – Part 3: Digital interfaces based on IEC 61850**
  - Test arrangement above

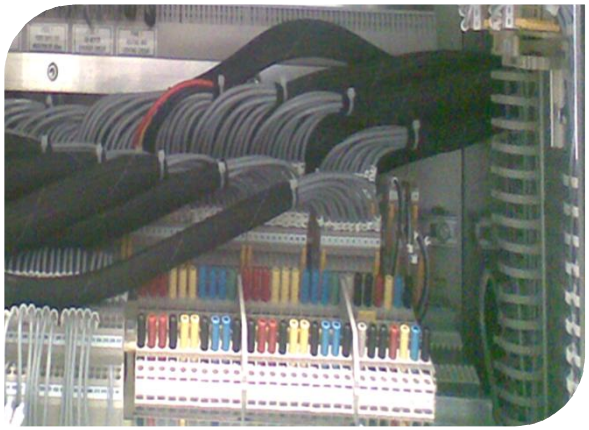
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Relion® Performance
Comparison between hardwired and GOOSE performance



- GOOSE performance is faster than the overall hardwired one
- Relion® IEDs fullfill standard requirement for protection applications
- GOOSE has higher error detection
  - Interruption of communication link is detected within configurable setting
- KEMA test report according *IEC 62271-3, High-voltage switchgear and control gear – Part 3: Digital interfaces based on IEC 61850* available

Relion® Performance
Utilizing GOOSE in Switchgear



Use case:

- UniGear with 10 bays
  - Load management system, as well as control system interface
- ~58% of the signals are between the relays or other automation
  - Replaced by Ethernet

Relion® Performance
Solution benefits with GOOSE



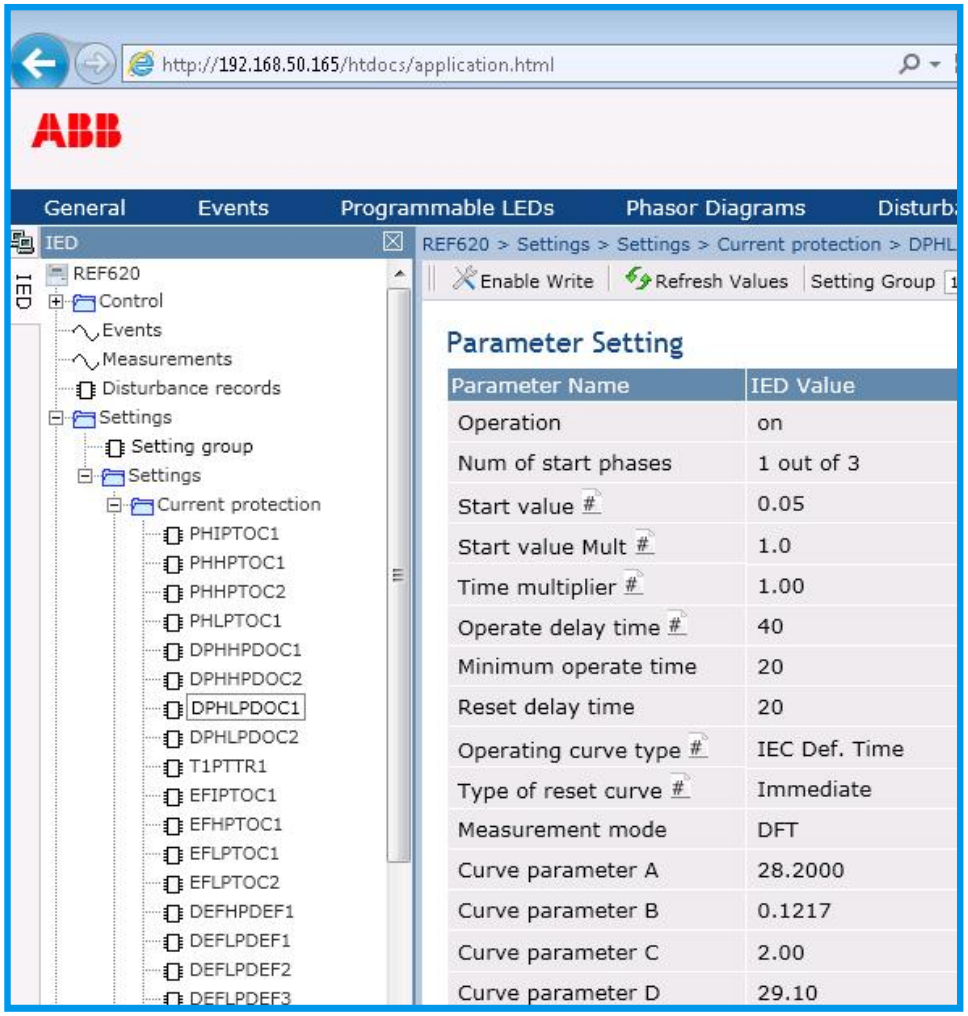
- Application changes can be easily implemented via software without being force to modify wiring and relevant drawings

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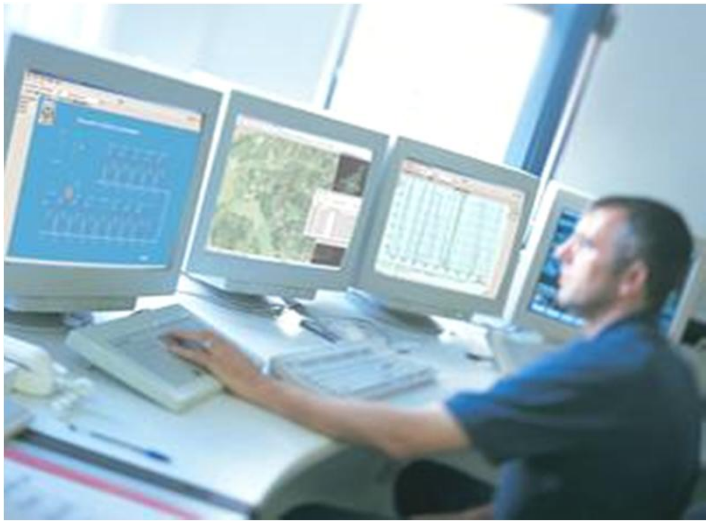
IEC 61850 in DA Products
Advanced Usage of Ethernet



- WebHMI
  - Viewing of alarm & event list
  - Saving of event data
  - Parameter setting
  - Signal monitoring
  - Measurement viewing
  - Phasor Diagram viewing
- Fault Records
  - Upload, analysis
- Disturbance Recordings
  - Upload, analysis
- IED Tool Connectivity
  - Local, Remote
- Service Bus access
  - Settings export/import

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IEC 61850 in DA Products
Relion®



IEC 61850 COMPLIANT

- Native support for IEC 61850 communication
- Native IEC 61850 enables fast “GOOSE” (Generic Object Oriented Substation Event) horizontal communication between IEDs
- KEMA IEC 61850 conformance statements available
- In 615/620 Ethernet redundancy as option
- 615 and 620 series also supports in parallel with IEC 61850:
  - Modbus TCP/IP and RTU/ASCII
  - DNP3 TCP/IP and serial
  - IEC 60870-5-103
- 630 series also supports:
  - DNP3 TCP/IP
  - IEC 60870-5-103

IEC 61850 in DA Products
615, 620 and 630 series IEC 61850

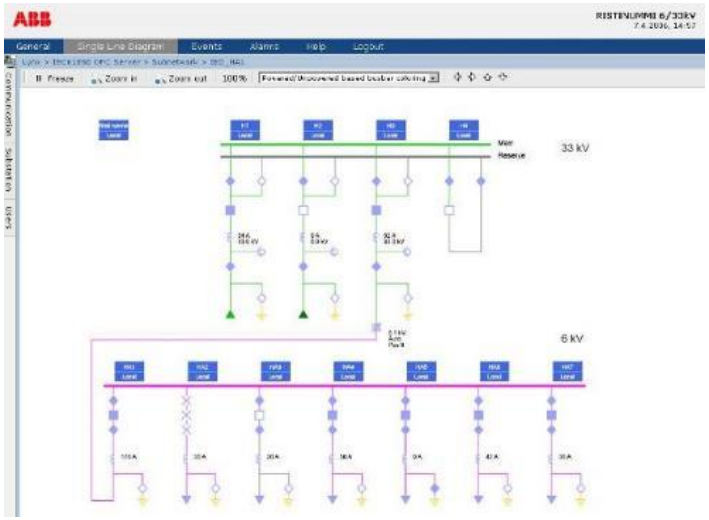


- IED core protocol for all functionality and applications
- Settings and parameterization are done fully according IEC 61850 services
- COMTRADE disturbance recordings over IEC 61850 or FTP
- Fault records available in IEC 61850
- Fast GOOSE messaging between IEDs fulfilling IEC 61850-5 requirement, data transmission time <10 ms from IED to IED
- SNTP for time synchronization with 1ms accuracy
- Support of five parallel clients

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IEC 61850 in DA Products
COM600





- Powerful gateway between substation and IEDs
- Takes full advantage of DA IEDs IEC 61850 features improving engineering efficiency
- Easy to use local or Web HMI with single line diagram, even&alarm list and much more
- Intelligent real-time applications in COM600 with 61131 Logic Processor
- Supports GOOSE e.g. with RIO600
- Cyber Security optimized

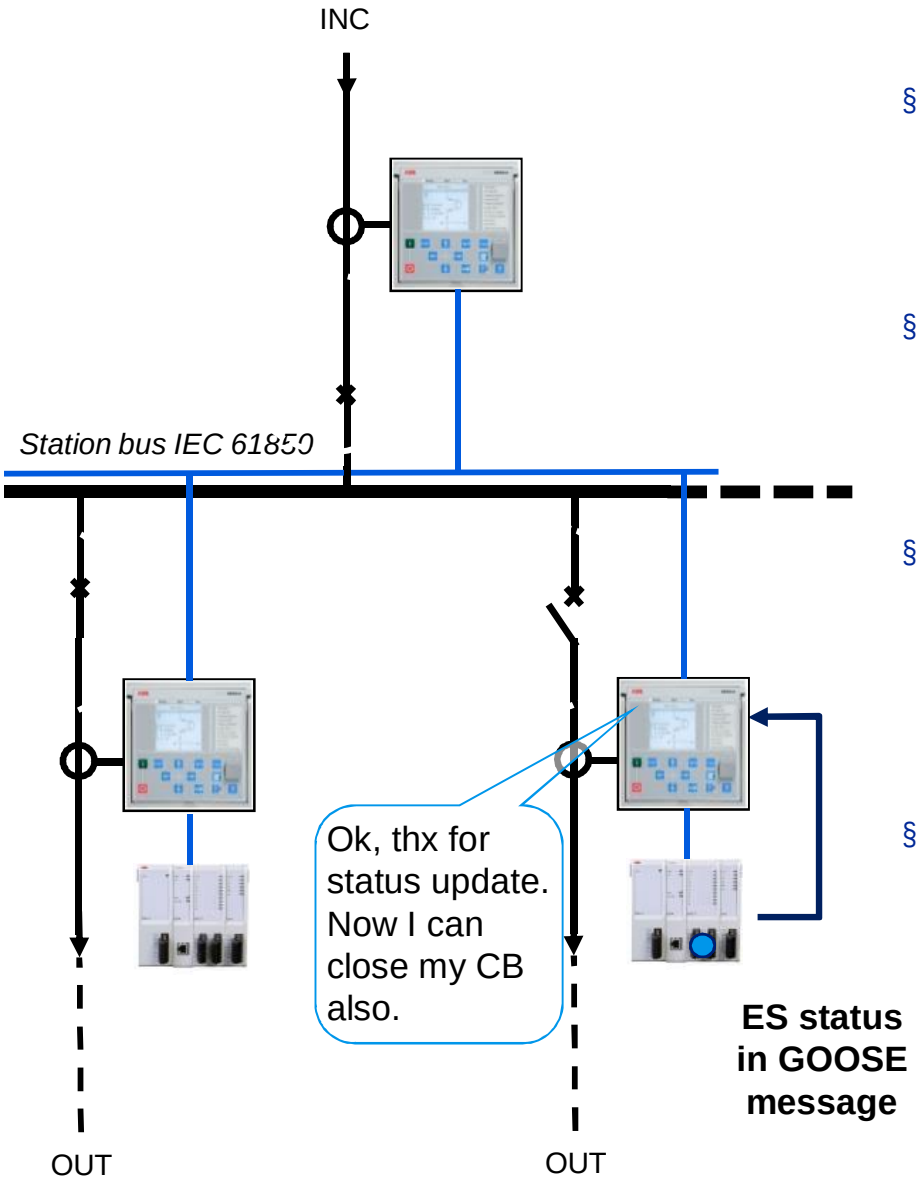
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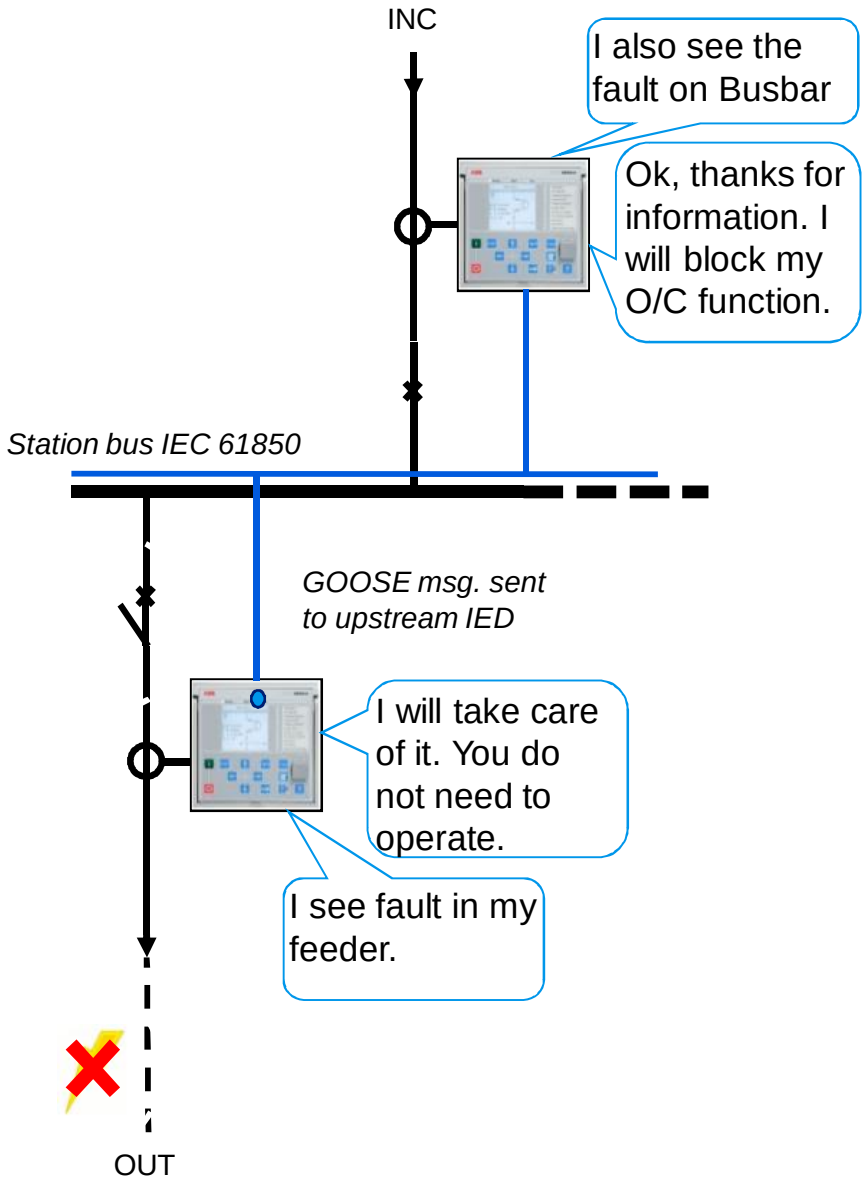
Application Examples
Remote I/O RIO600



- RIO600 inputs are connected to Earthing Switch open and closed statuses
- In case there is change of status RIO600 sends information to IED B
- Due change in station interlocking scheme IED B sends notification to other IEDs in IEC 61850 station bus
- The use of RIO600 extends the available I/O space for Relion® IEDs

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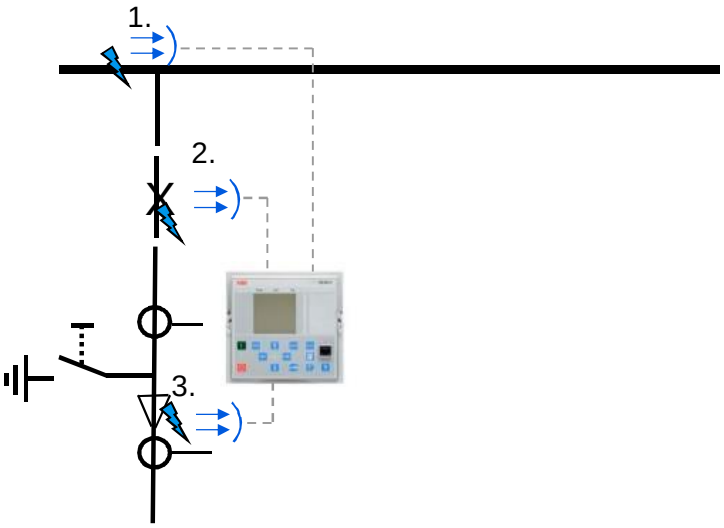
Binary GOOSE Application
Busbar Protection (O/C blocking)



- GOOSE can be used for Logic Busbar Protection (Overcurrent blocking protection) for logic protection coordination
- Fault on cable side on outgoing feeder is seen by IED in the OUT feeder and by IED in incoming feeder, due to the set parameters for Overcurrent protection function (I>>)
- I>>> Start from outgoing feeder is shared via GOOSE to incoming feeder where it blocks the I>> function
- Fault is cleared out by Circuit Breaker in outgoing feeder faster than hard wired

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Application examples
Arc protection using REF615





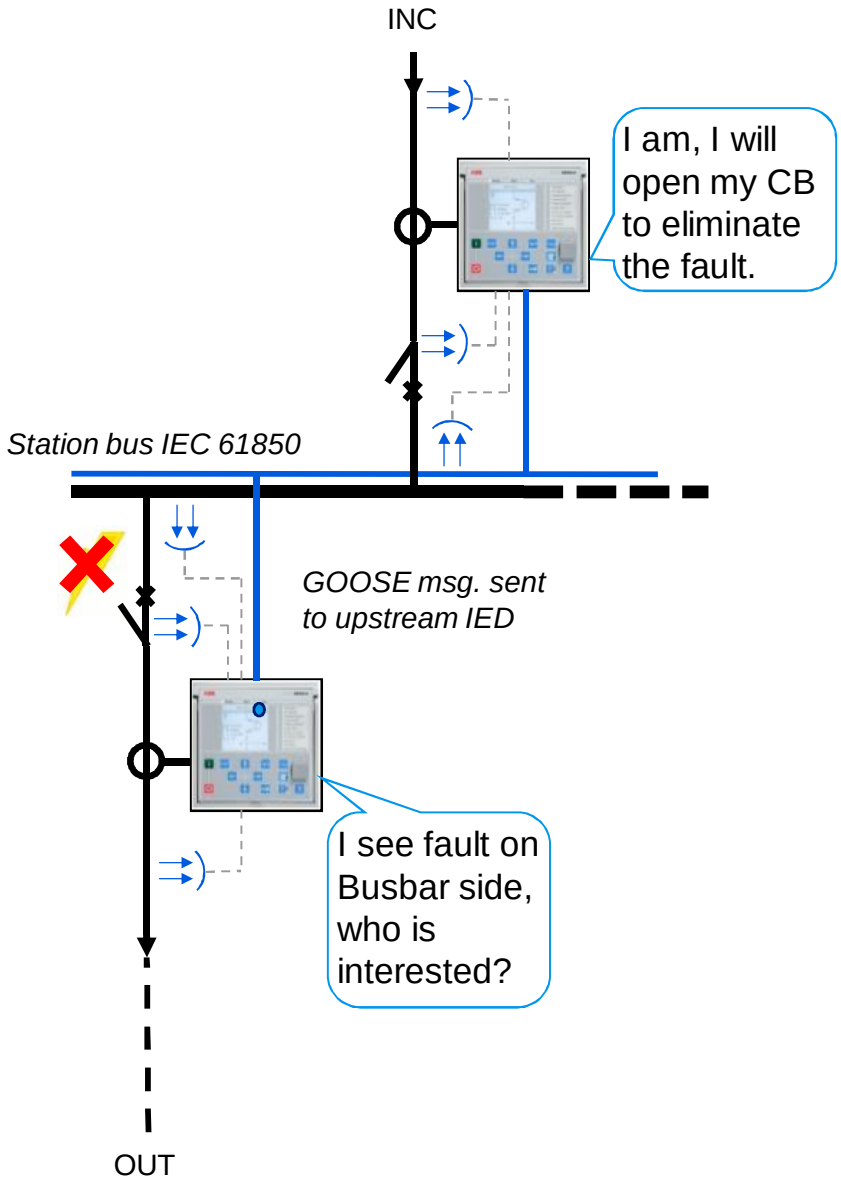
Three separate sensors

- Busbar compartment
- Circuit-breaker compartment
- Cable compartment

Operating conditions

- Arc detected
- Arc and current detected
  - Current threshold separately adjustable for phase currents and neutral current

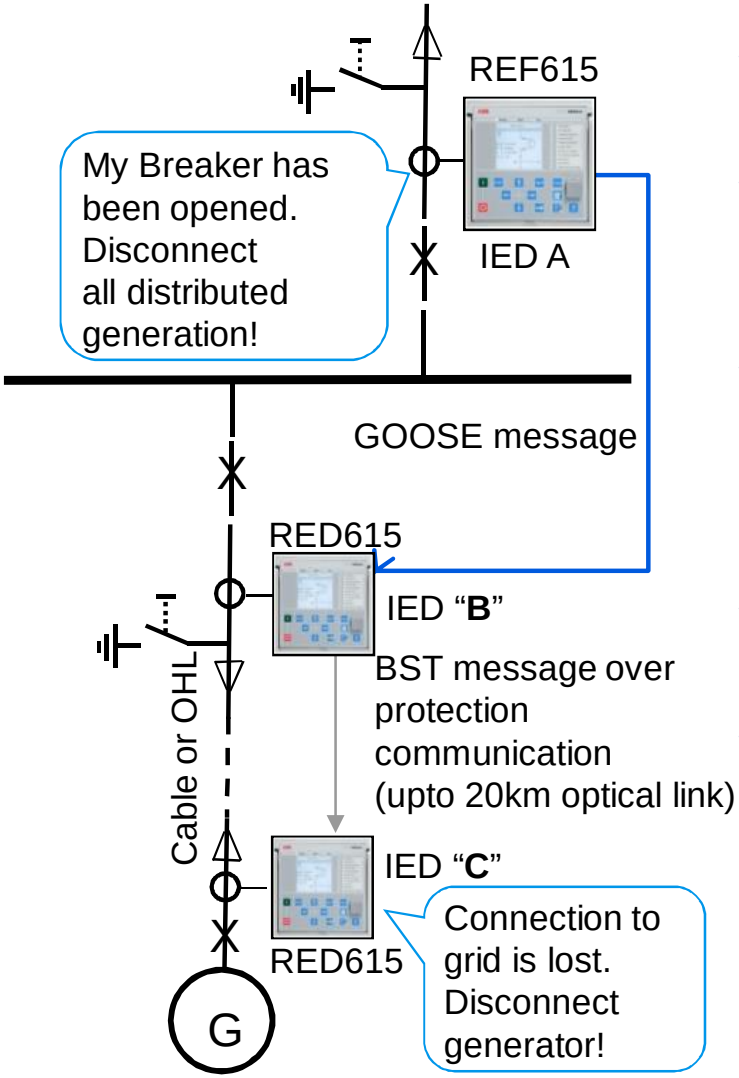
Binary GOOSE Application
ARC Protection Intertripping



- GOOSE can be used for ARC protection intertripping within the switchgear
- IED in OUT feeder detects the ARC e.g. on busbar side
- The IED in INC have to eliminate this fault
- IED in INC will open the CB and fault is cleared

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Application examples
Loss-of-mains protection using GOOSE and BST

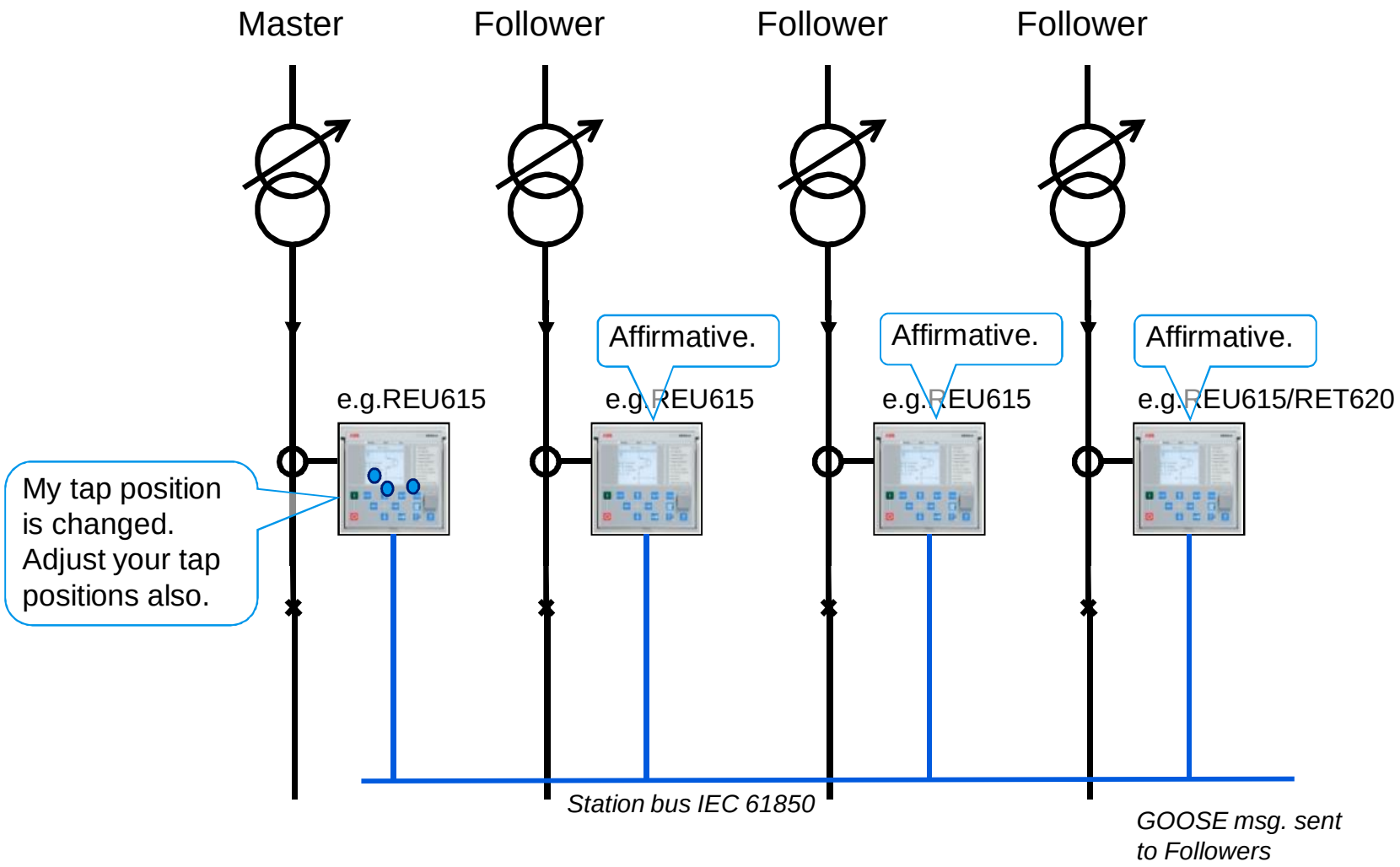


- IED A on the incoming feeder detects an open circuit-breaker condition
- The CB position information is transferred to Relay B as a GOOSE message
- After receiving the GOOSE message Relay B transfers the information to Relay C using BST, binary signal transfer
- Relay C will issue a command to disconnect the generator from network
- Whole communication chain 28ms!

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KM2

Binary GOOSE application
Tap changer control with AVR (Master/Follower mode)

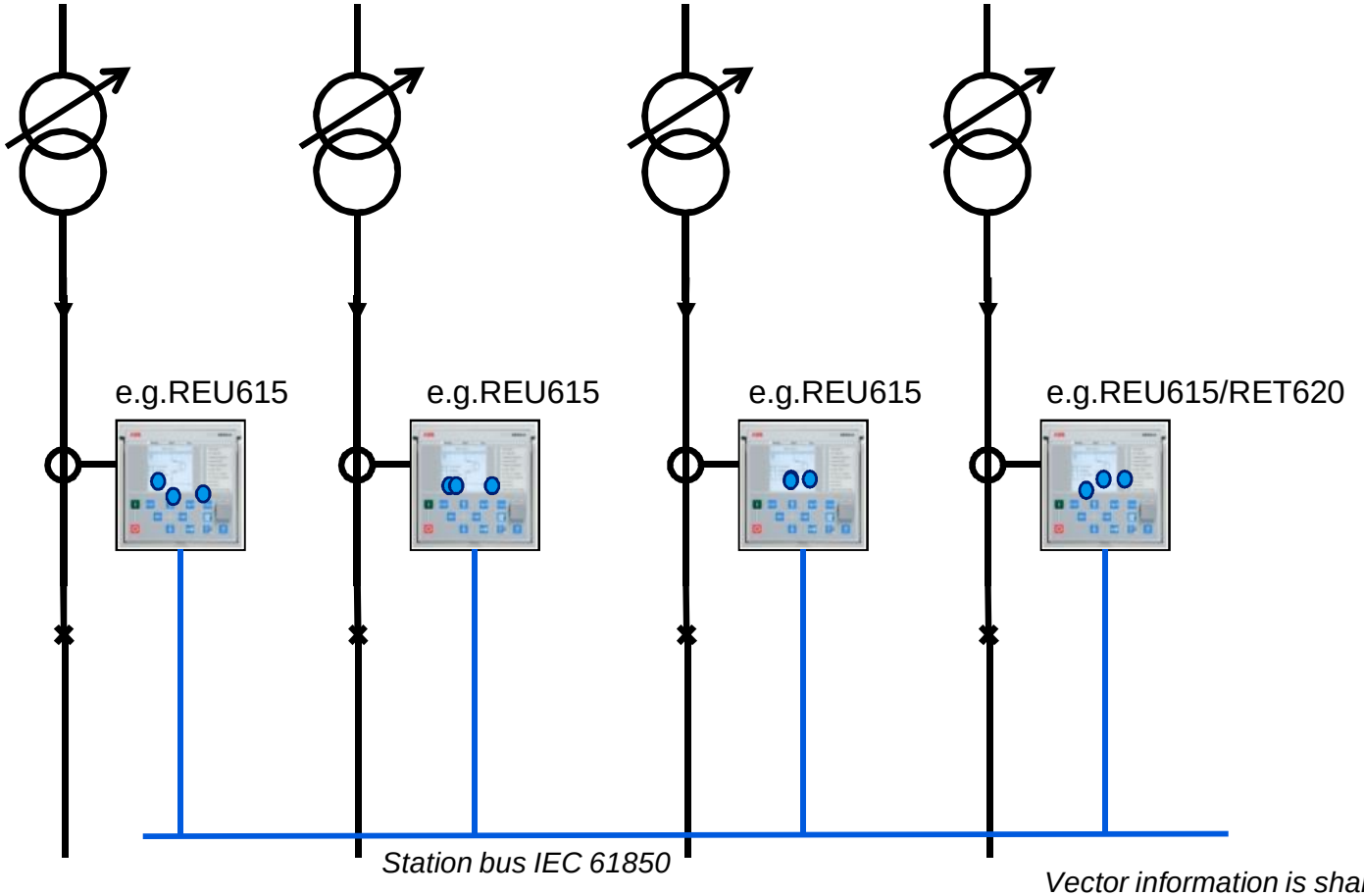


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KM2 tento tiez zanimovat nejak

Karol Majer; 22.5.2014

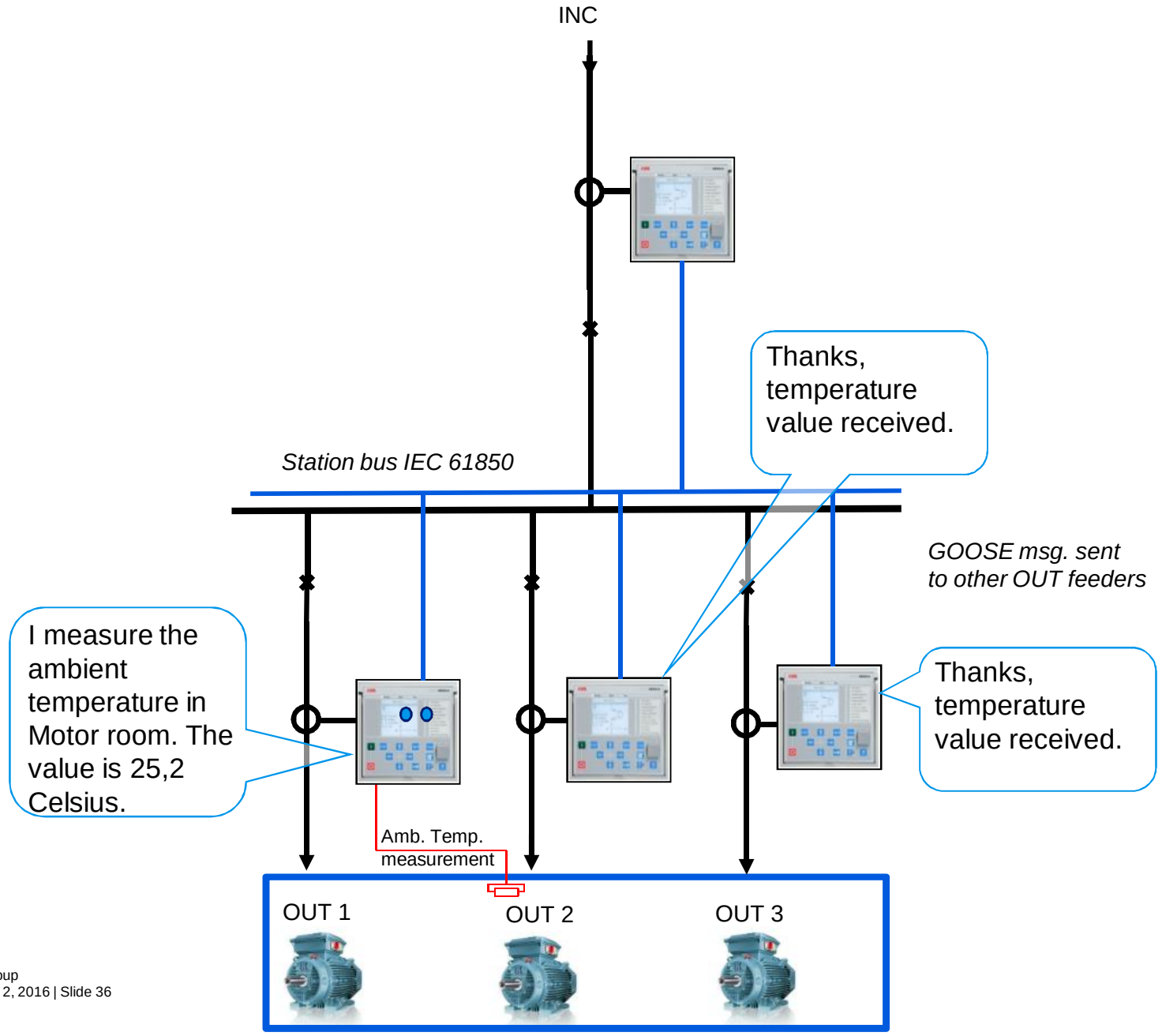
Analog GOOSE application
Tap changer control with AVR (Minizing circulating current mode)



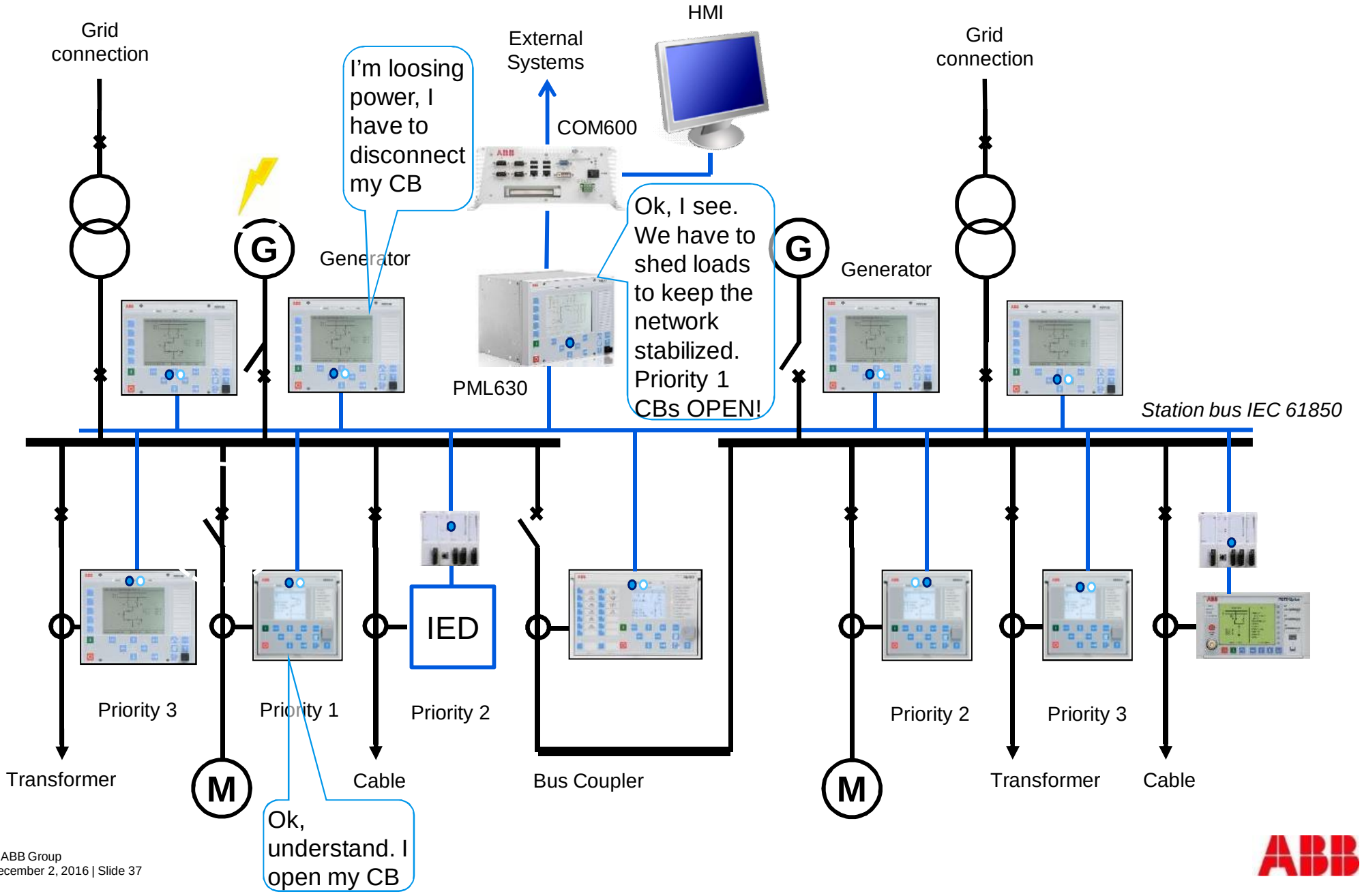
Vector information is shared over Analog GOOSE between regulators

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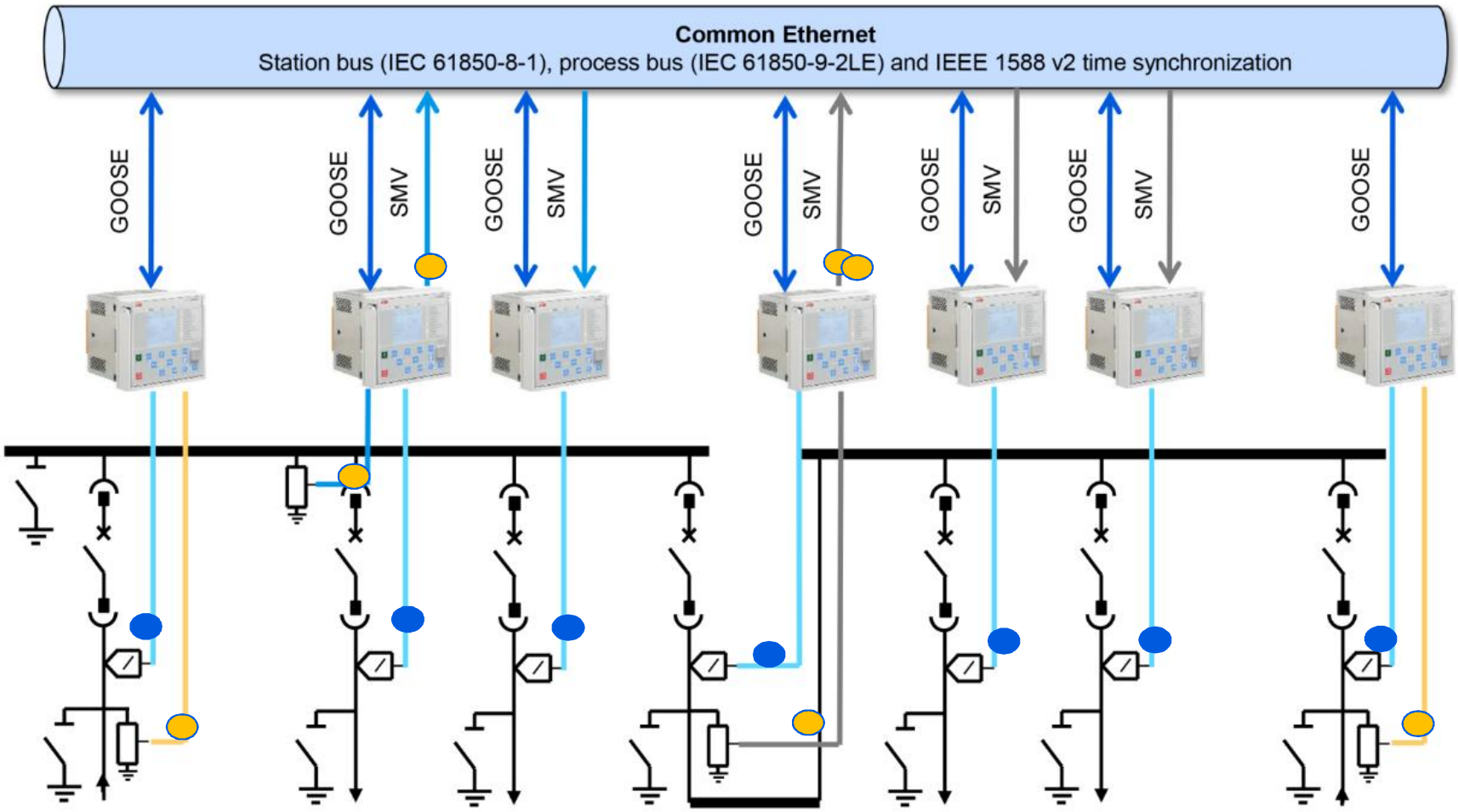
Analog GOOSE application
Temperature measurement in Motor substation



Binary and Analog GOOSE application
“Active” Load-shedding with PML630



IEC 61850 9-2 Sampled Values
Process Bus Example



Raw phase measurements sent with 4kHz sampling rate

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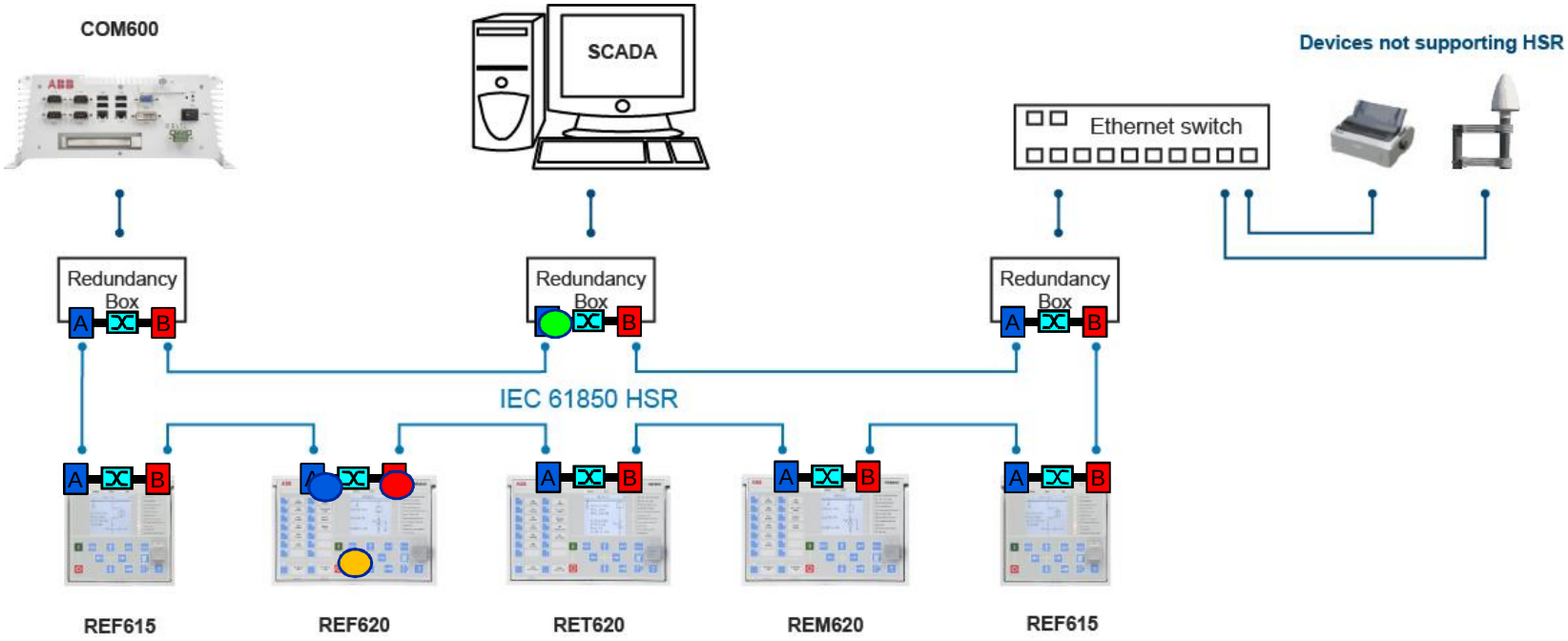
Redundancy solutions retained for IEC 61850

- With “zero” switchover time (seamless)
  - **PRP** (Parallel Redundancy Protocol)
    IEC 62439-3 Clause 4
  - **HSR** (High-availability Seamless Redundancy)
    IEC 62439-3 Clause 5
- PRP/HSR is a patent of ABB Corporate Research
- ABB has assured the IEC to provide licenses throughout the world free of charge or under reasonable terms

HSR
Normal operation

Operation description:

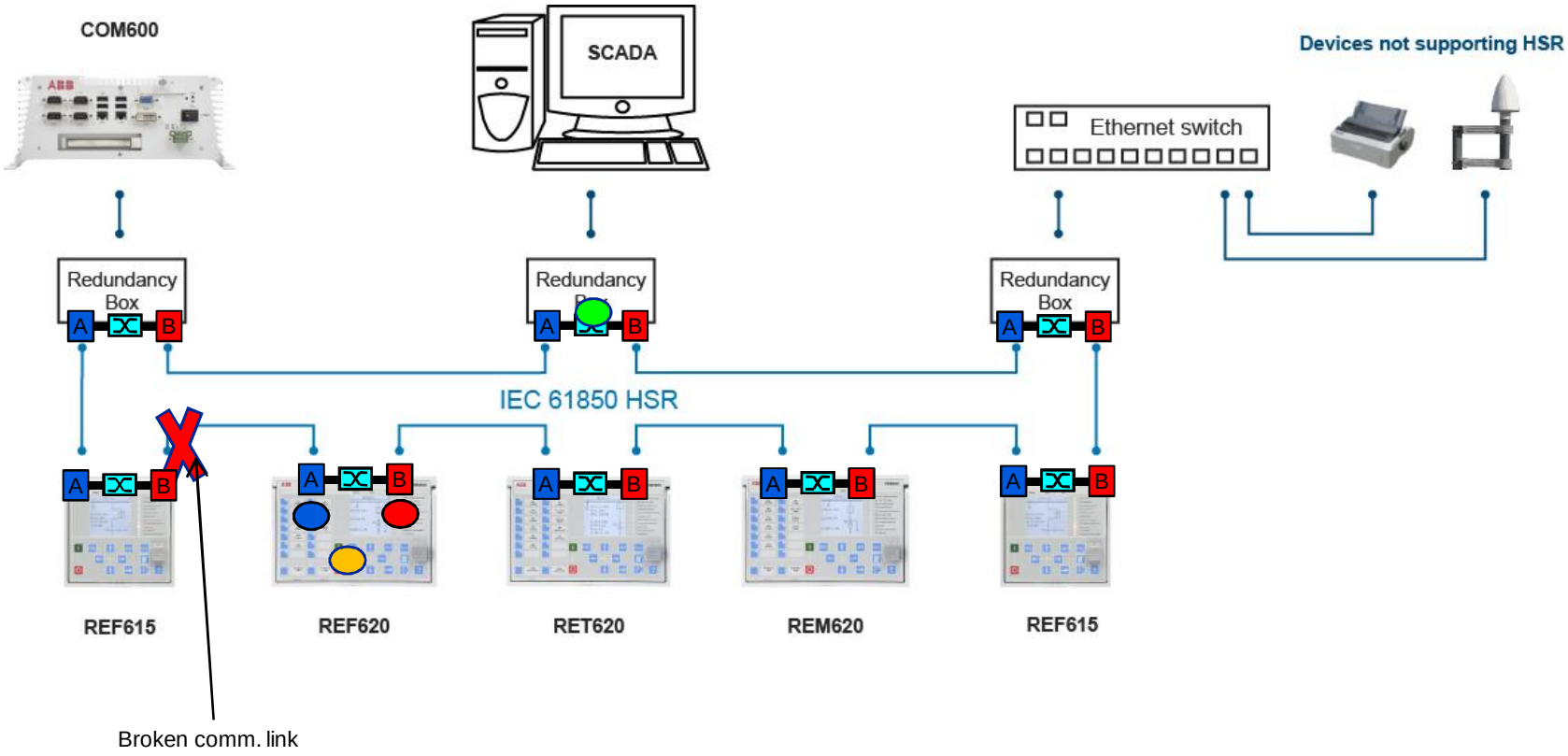
1. Message from IED is sent via both links (“A” and “B”) to the SCADA via HSR ring



HSR
Operation under failure condition

Operation description:

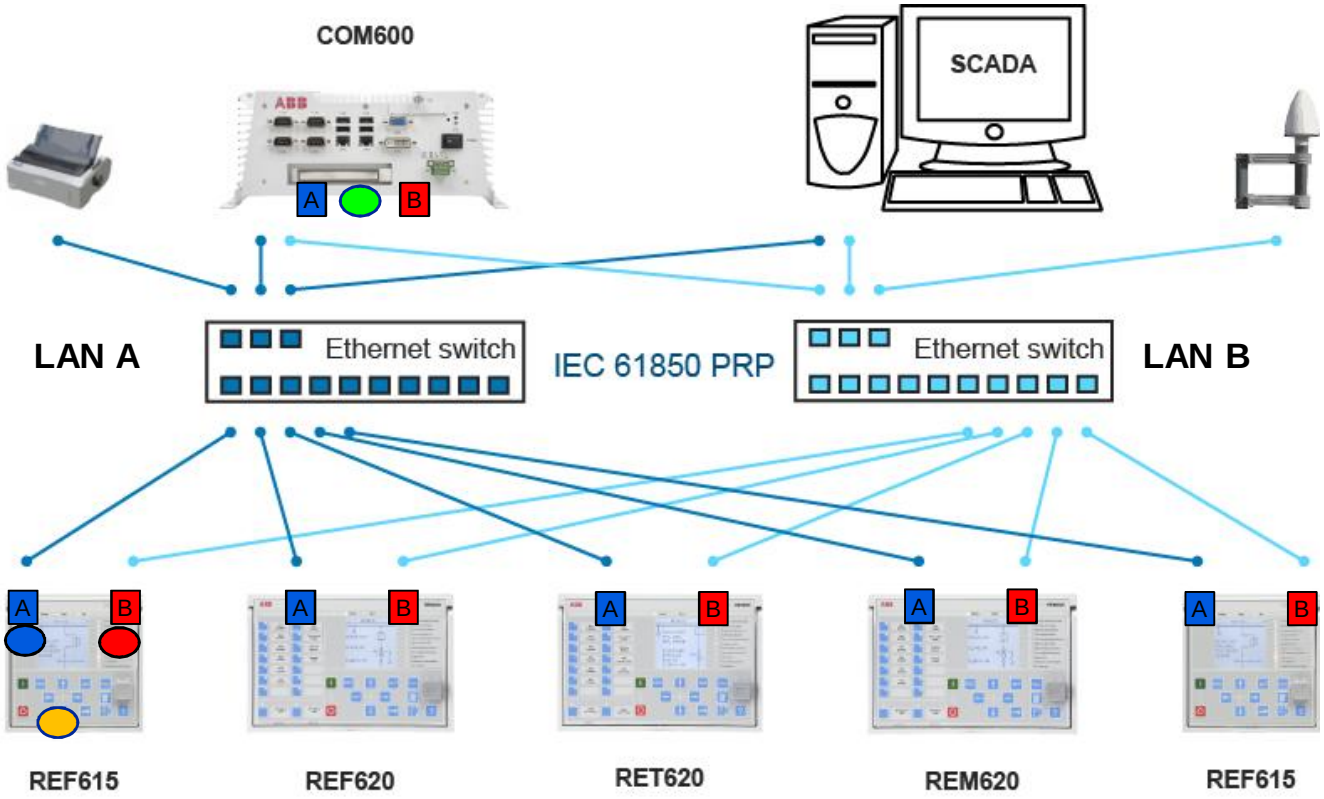
1. Failure recognized in HSR ring (“A” link)
2. Message is received by SCADA via healthy part of ring (“B” link)



PRP
Normal operation

Operation description:

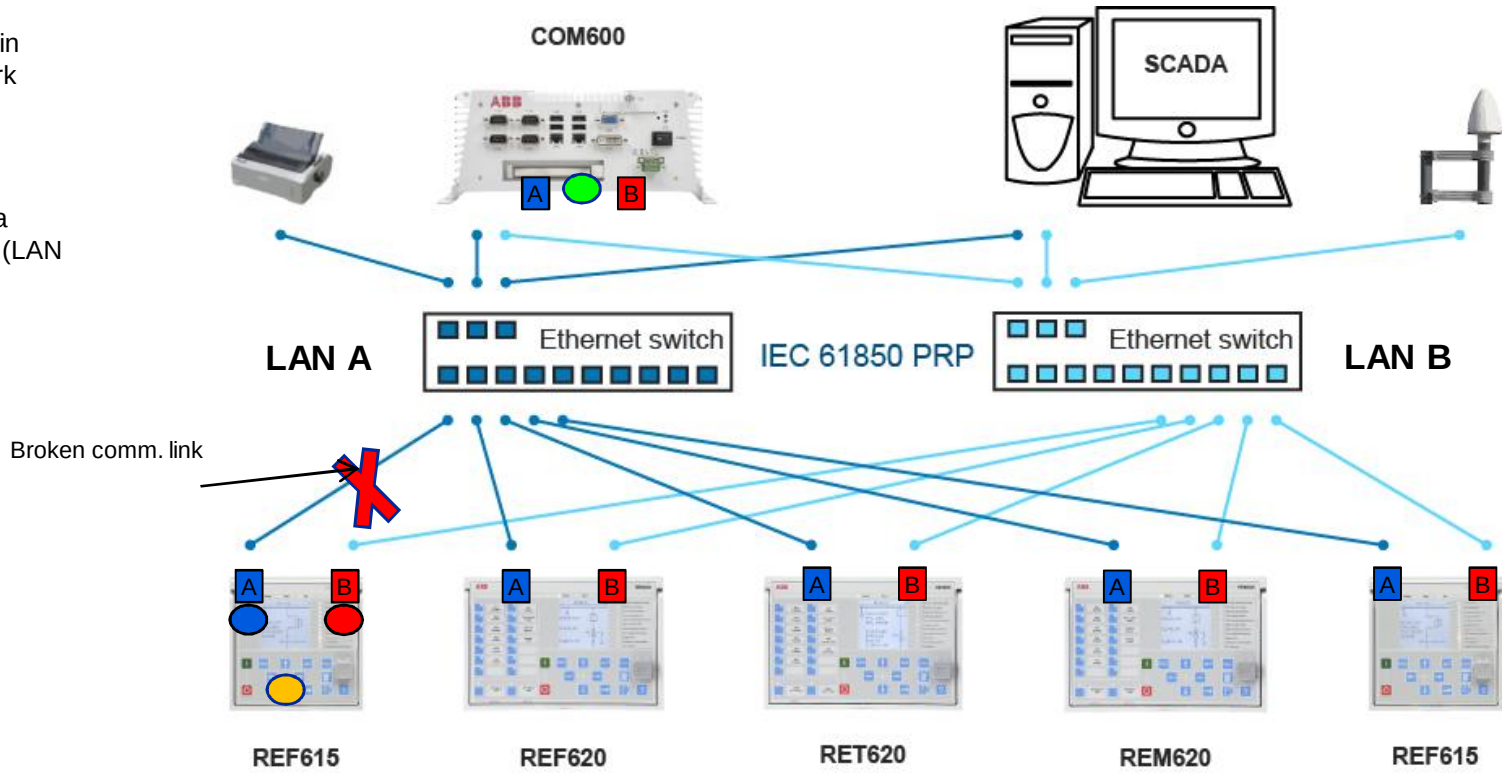
1. Message is received in COM600 via both parallel links (LAN “A” and LAN “B”)



PRP
Operation under failure condition

Operation description:

1. Failure recognized in PRP network (LAN “A”)
2. Message is received in COM600 via healthy link (LAN “B”)



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Communication Solutions
Conclusions

- IEC 61850 offers **system integration and interoperability**
- **Less wiring** between IEDs means less equipment and work
- GOOSE and 9-2 has **superior functionality** comparing to traditional wiring (performance, link and data supervision)
- **Flexible utilization of available I/Os** in the substation equipment
- Substation automation and IED configuration is more **intelligent and flexible**
- Seamless **Redundant Ethernet** provides improved availability
- **Future proof** communication standard choice

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